1. Find the Co-ordinates of the point, which divides the line segments joining (2, 0) and (0,2) in the ratio 1 : 1

Solution:

$$\frac{m_1x_2+m_2x_1}{m_1+m_2},\frac{m_1y_2+m_2y_1}{m_1+m_2}$$

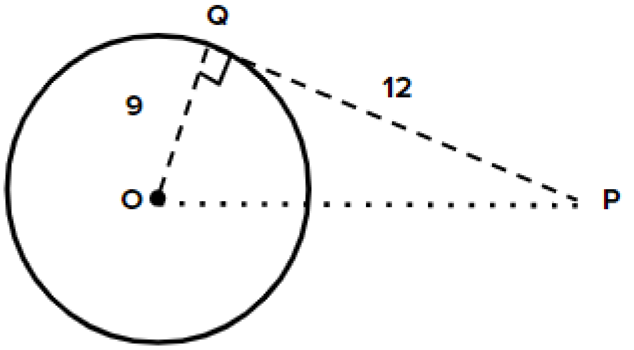
$$=\frac{1\times 0+1\times 2}{1+1},\frac{1\times 2+1\times 0}{1+1}$$

$$=\frac{2}{2},\frac{2}{2}$$

$$=1,1$$

2. 'O' is the centre of a circle. PQ is a tangent to the circle at Q from the external point P. If the radius of circle is 9 cm and PQ = 12 cm, find the distance of P from O.

Solution:



In Δ PQO

$$PO^2 = PQ^2 + OQ^2$$

$$=9^2+12^2$$

$$=81+144$$

$$PO=\sqrt{225}$$

$$= 15\text{ CM}$$

3. Find the value of x, if 2sinx = $\sqrt{3}$.

Solution:

2sinx = $\sqrt{3}$

sin x = $\frac{\sqrt{3}}{2}$

sin x = sin 60

x = 60

4. You are writing a test of 40 objective type questions. Each questions carries 1 mark. What is the probability of marks you may get to be in multiple of 5?

Solution:

Event = {5, 10, 15, 20, 25, 30, 35, 40}

n(A) = 40

$p = \frac{8}{40} = \frac{1}{5}$

5. Find the value of k, for which the points (7, 2), (5, 1) and (3, k) are collinear.

Solution:

A = 0

$$\frac{1}{2}\{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\} = 0$$

$$\frac{1}{2}\{7(1-k) + 5(k-2) + 3(2-1)\} = 0$$

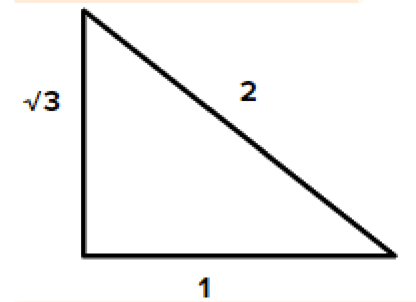
$$7 - 7k + 5k - 10 + 3 = 0$$

$$-2k = 0$$

$$k = 0$$

6. Find $\angle B$, if tan(A- B) = $\frac{1}{\sqrt{3}}$ and sin A = $\frac{\sqrt{3}}{2}$. Also find cos B. (A, B < 90°)

Solution:



Sin A = $\frac{\sqrt{3}}{2}$

Sin A = sin 60

A = 60

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3} - \tan B}{1 + \sqrt{3}\tan B}$$

$1 + \sqrt{3}\tan B = 3 - \sqrt{3}\tan B$

$2\sqrt{3}\tan B = 3 - 1$

$\tan B = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$

tan B = tan 30

B =30°

Then, cos 30 = $\frac{\sqrt{3}}{2}$

7. Give two different example of pair of
(i) Similar figures
(ii) Non-similar figures.

Solution:

(i) Two triangles
(ii) A square and A triangle

8. There are 5 cards in a box with numbers 1 to 5 written on them. If 2 cards are picked out from the box, write all the possible outcomes and find the probability of getting both even numbers.

Solution:

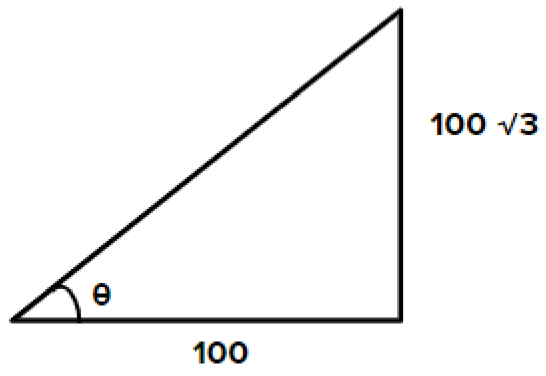
n(s) = {(1, 1) (1, 2) (1, 3) (1, 4) (1, 5) (2, 1) (2, 2) (2, 3) (2, 3) (2, 4) (2, 5) (3, 1) (3, 2) (3, 3) (3, 4) (3, 5) (4, 1) (4, 2) (4, 3) (4, 4) (4, 5) (5, 1) (5, 2) (5, 3) (5, 4) (5, 5)}

Event events = {(2, 2) (2, 4) (4, 2) (4, 4)}

$$P(E) = \frac{4}{25}$$

9. A tower is 100 $\sqrt{3}$ in high. Find the angle of elevation of its top when observed from a point 100m away from the foot of the tower.

Solution:



$$\tan\theta = \frac{100\sqrt{3}}{100} = \sqrt{3}$$
$$\theta = 60^\circ$$

10. (a) A wire of length 18 cm had been tied to an electric pole at angle of elevation 30° with the ground. As it is covering a long distance, it was cut and tied to the pole at an angle of 60° with the ground. Now, find how much length of the wire was cut?

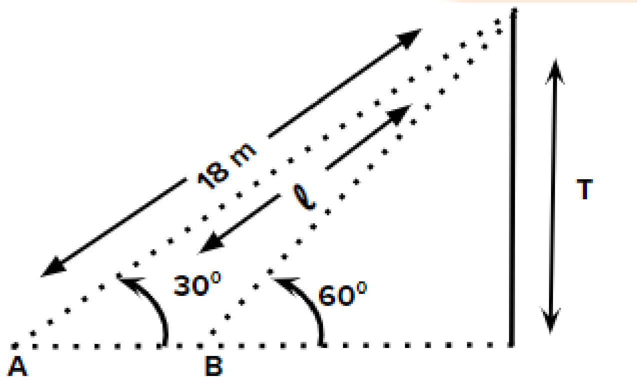
(b) Consider the following distribution of daily wages of 50 workers of a factory.

Daily Wages (in Rs)	200-250	250-300	300-350	350-400	400-450
No.of Workers	12	14	8	6	10

Find the mean daily wages of the workers by choosing an appropriate method.

Solution:

10(a)



Sin 30 = $\frac{T}{18}$

$T = \frac{1}{2} \times 18 = 9m$

$\sin 60° = \frac{T}{l_1}$

$l_1 = \frac{2}{\sqrt{3}} \times 9$

l = 10.99 m

10(b)

wage	marks	No. x_i	a x_i
200-250	225	12	2700
250-300	275	14	3850
300-350	325	8	2600
350-400	375	6	2250
400-450	425	10	4250
		Σx_i =50	$\Sigma a_i x_i$ = 15650

Mean = $\frac{\sum a_i x_i}{\sum x_i} = \frac{15650}{50}$

= 313

11. (a) Prove that

$$(\sin\theta-\operatorname{cosec}\theta)^2+(\cos\theta-\sec\theta)^2 = \cot^2\theta+\tan^2\theta-1$$

(b) Check whether the points (3,0), (6,4) and (-1, 3) are the vertices of a right- angle isosceles triangle or not. Also find the area of the triangle.

Solution:

11(a)

$$(\sin\theta-\operatorname{cosec}\theta)^2+(\cos\theta-\sec\theta)^2$$

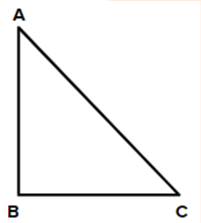
$$\sin^2\theta+\operatorname{cosec}^2\theta-2\sin\theta\operatorname{cosec}\theta+\cos^2\theta+\sec^2\theta-2\cos\theta\sec\theta$$

$$(\sin^2\theta+\cos^2\theta)+\operatorname{cosec}^2\theta-2+\sec^2\theta-2$$

$$1+1+\cot^2\theta-4+1+\tan^2\theta$$

$$\cot^2\theta+\tan^2\theta-1=\text{R.H.S}$$

11(b)



Let A = 3, 0 B = 6, 4 C = -1, 3

$$AB=\sqrt{(6-3)^2+(4-0)^2}=\sqrt{9+16}=5\text{ cm}$$

$$BC=\sqrt{(-1-6)^2+(3-4)^2}=\sqrt{49+1}=\sqrt{50}\text{ cm}$$

$$CA=\sqrt{(-1-3)^2+(3-0)^2}=\sqrt{16+9}=5\text{cm}$$

So, $\because$ AB = CA (they are Isosceles)

$\because$ $AB^2+CA^2=BC^2$ (Hence ABC is right angle)

$$\text{Area}=\frac{1}{2}b\times h=\frac{1}{2}\times5\times5=\frac{25}{2}\text{cm}^2$$

12. (a) A chord of circle of radius 10 cm subtends a right angle at the centre.

Find the area of the corresponding:

(i) Minor segment.

(ii) Major segment.

(use π = 3.14)

(b) From a deck of 52 playing cards, King, Ace and 10 of clubs were removed are remaining cards were well shuffled. If a card is drawn at random from the remaining, find the probability of getting a card of

(i) Club

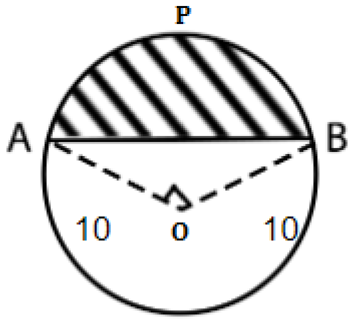
(ii) Ace

(iii) Diamond King

(iv) Club 5.

Solution:

12(a)



Given that OA = OB = 10 cm

$\theta = 90°$

Area of segment APB = Area of sector

OAPB = Area of Δ AOB

Area of sector OAPB = $\dfrac{\theta}{360} \times \pi^2$

$= \dfrac{90}{360} \times \dfrac{22}{7} \times 10 \times 10$

$78.5\,\text{cm}^2$

Area of Δ AOB = $\dfrac{1}{2}$bh

$= \dfrac{1}{2} \times OA \times OB = \dfrac{1}{2} \times 10 \times 10 = 50\ \text{cm}^2$

Area of segment OAPB = 78.5 – 50 = 28.5 cm^2

Major segment Area = Area of circle – Area of minor segment

$= \pi r^2 - 28.5$

$= \; = 3.14 \times 100 - 28.5$

$= 285.6\,\text{cm}^2$

12(b)

n(s) = 52 – 4 – 4 – 1 = 43

(i) n(Club) = 10

p = $\dfrac{10}{43}$

(ii) A(Ace) = 0 (not present)

p = 0

(iii) n(Diamond King) = (Already removed)

p = 0

(iv) n(club 5) = 1

p = $\dfrac{1}{43}$

13. (a) Draw a circle of radius 3 cm. Take a point 'P' at a distance of 5 cm from the centre of the circle. From P, draw 2 tangent to the circle.

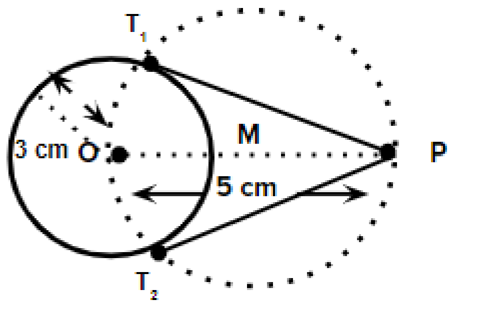
(or)

(b) Draw "greater than Ogive curve" for the following data.

Classes	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	4	4	8	10	12	8	4

Solution:

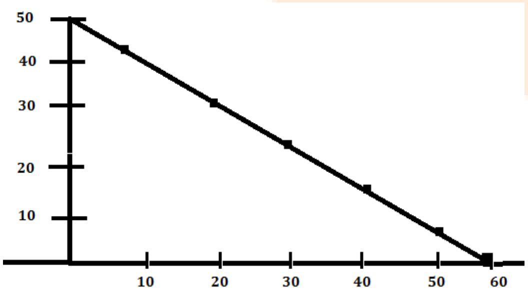
(a)



1) Draw circle of 3 cm
2) Take p at 5 cm from 0
3) Draw circle of 5 cm with centre M.
4) Whatever it cuts name it as T_1 and T_2 and join.

(b)

Classes	**f**	**Cf(>)**
0 – 10	4	50
10 – 20	4	46
20 – 30	8	38
30 – 40	10	28
40 – 50	12	16
50 – 60	8	8
60 – 70	4	4



14. If origin is the centroid of a triangle, whose vertices are (3, 2), (-6, y), (3, -2), then y =

Options:

(a) 0

(b) 3

(c) 2

(d) 6

Answer: (a)

Solution:

$$\frac{3+(-6)+3}{3}, \frac{2+y-2}{3} = (0,0)$$

$$0 = \frac{2+y-2}{3}$$

0 = y+0

Y = 0

15. Areas of 2 similar triangles are 100 cm^2 and 64 cm^2 If the median of bigger triangle is 10 cm, then the median of the smaller triangle is …..

Options:

(a) 10 cm

(b) 6 cm

(c) 4 cm

(d) 8 cm

Answer: (d)

Solution:

$$\frac{Ar.of\,1}{Ar.of\,2} = \left(\frac{median\,1}{median\,2}\right)^2$$

$$\frac{100}{64} = \left(\frac{10}{x}\right)^2$$

$$\frac{10}{8} = \frac{10}{x}$$

x = 8 cm

16. If sin x = $\frac{5}{7}$**,** then cosec x = …….

Options:

(a) $\frac{5}{7}$

(b) $\frac{7}{5}$

(c) $\frac{2}{5}$

(d) $\frac{2}{7}$

Answer: (b)

17. Given $\angle A = 75°, \angle B = 30°,$ then tan(A - B) = …..

Options:

(a) $\sqrt{3}$

(b) $\frac{1}{\sqrt{3}}$

(c) 1

(d) $\frac{1}{\sqrt{2}}$

Answer: (c)

Solution:

tan (A -B) = tan (75 – 30)

= tan 45 = 1

18. If P(E) = 0.26, than $P(\overline{E})$ = ……

Options:

(a) 0.74

(b) 0

(c) 0.26

(d) 1

Answer: (a)

Solution:

1 – 0.26 = $p(\overline{E})$

0.74

19. Median of 2, 3, 4, 5, 6, 7 is …….

Options:

(a) 2

(b) 5.5

(c) 5

(d) 4.5

Answer: (d)

Solution:

$$\text{Median} = \frac{\left(\frac{n+1}{2}\right)^{th} + \left(\frac{n}{2}\right)^{th}}{2} = \frac{4+5}{2} = 4.5$$

20. Which of the following cannot be a point on X-axis?

Options:

(a) (-2, 0)
(b) (0, 2)
(c) (2, 0)
(d) (4, 0)
Answer: (b**)**

21. Radius of a circle with centre 'O' is 5 cm. P is point at a distance of 3 cm from 'O'. Then the number of tangents that can be drawn to the circle is ……
Options:
(a) 1
(b) 2
(c) 0
(d) 3
Answer: (c**)**
Solution:
Because p point is still inside the circle

22. If $\sec\theta-\tan\theta=\frac{1}{3}$, then $\sec\theta+\tan\theta=$ …….
Options:
(a) 3
(b) $\frac{1}{3}$
(c) 1
(d) 0
Answer: (a**)**
Solution:

$$\sec\theta+\tan\theta=\frac{1}{3}$$

$$\frac{1+\sin\theta}{\cos\theta}=\frac{1}{3}$$

$$\sec^2\theta-\tan^2\theta=1$$

$$(\sec\theta+tan\theta)(\sec\theta-\tan\theta)=1$$

$$\sec\theta-\tan\theta=\frac{1}{\frac{1}{3}}=3$$

23. Probability of getting 7, when dice is rolled, is ……
Options:
(a) $\frac{1}{6}$
(b) $\frac{1}{7}$

(c) $\frac{6}{7}$

(d) 0

Answer: (d)

Solution: 7 is not present on dice

24. To elect the leader of your class from 3 contestants, which of the following measures are to be considered?

Options:

(a) Mean

(b) Mode

(c) Median

(d) Range

Answer: (b)

Solution:

Most favourite leader or any from is always formed as mode.

25. In Heron's formula, area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$ s is …. Of the triangle.

Options:

(a) Perimeter.

(b) Height

(c) Half of perimeter.

(d) None

Answer: (c)

26. Angle made by the minutes-hand in a clock during a period of 20 minutes is ……

Options:

(a) 120°

(b) 20°

(c) 360°

(d) 90°

Answer: (a)

Solution:

For 60 minutes $\rightarrow 360°$

$$1 \rightarrow \frac{360}{60}$$

$$20 \rightarrow \frac{360}{60} \times 20 = 120°$$

27. Which of the following situations have equally likely events?

(1) Getting 1 or 2 or 3 or 4 or 5 or 6 when a dice is rolled.

(2) Winning or loosing a game.

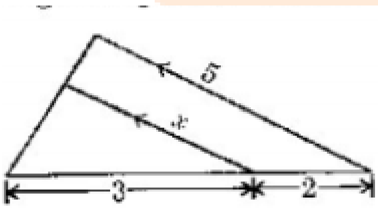
(3) Head or Tail, when a coin is tossed.

Options:
(a) 1 and 2
(b) 2 and 3
(c) 1 and 3
(d) All
Answer: (d)

28. The probability of picking a letter from the set of English alphabets is $\frac{5}{26}$. That alphabet can be …..
Because n(s) of vowel = 5 .
Options:
(a) consonant
(b) vowel
(c) any alphabet
(d) none
Answer: (b)

29. If $\Delta PQR \sim \Delta XYZ$ and $\angle X = 30°, \angle Q = 50°,$ then $\angle Z$ = ……
Options:
(a) 100°
(b) $\angle R$
(c) both A and B.
(d) not Known.
Answer: (b)
Solutions:
$\angle R$
$\because \Delta PQR \sim \Delta XYZ$
$so, \angle P = \angle X, \angle Q = \angle Y, \angle R = \angle Z$

30. From the given figure, x = ……..



Options:
(a) 3
(b) 2
(c) 5
(d) 1
Answer: (a)
Solution:

$\frac{3}{5} = \frac{x}{5}$

$\frac{3 \times 5}{5} = x$

$x = 3$ cm

31. Which of the following is the point of intersection of X- axis and the line y = x + 5?

Options:

(a) (0, 5)

(b) (5, 0)

(c) (0, -5)

(d) (-5, 0)

Answer: (d**)**

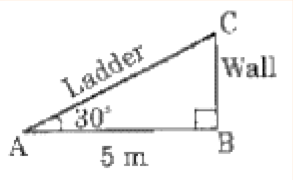
Solution:

x intercept = y=0

0 = x + 5

x = -5

32. Observe the figure. Length of the ladder = …….



Solutions: From figure,

cos 30 = $\frac{AB}{AC}$

AC = $\frac{5}{\sqrt{3}} \times 2 = \frac{10}{\sqrt{3}} = 5.77$

= 5 cm **(approx..)**

Options:

(a) 5 cm

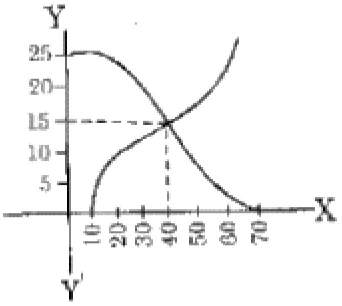
(b) 10 cm

(c) 20 cm

(d) 2.5 cm

Answer: (a**)**

33. From the given graph of O gives, median is …….



Options:
(a) 15
(b) 10
(c) 40
(d) 20
Answer: (a**)**
Solution:
Intersection of O give gives idea of median.